

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Artificial Intelligence Application Development and Jira

PRACTICAL 7

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Class: TYIT	Batch: 1
Date of Assignment: 08-08-2026	Date/Time of Submission: 08-08-2026

4. Implement k-fold cross-validation on a classification model.

Code:-

```
import pandas as pd
from sklearn.model_selection import cross_val_score
from sklearn.neighbors import KNeighborsClassifier
iris = pd.read_csv("iris - iris - iris - iris.csv")
X = iris.data
y = iris.target
model = KNeighborsClassifier()
scores = cross_val_score(model, X, y, cv=5)
print("Accuracy scores for each fold:")
print(scores)
print("Ritesh Chaubey T005")
print("Average Accuracy:", round(scores.mean() * 100, 2), "%")
```

Output:-

```
Accuracy scores for each fold:
[0.96666667 1.          0.93333333 0.96666667 1.          ]
Ritesh Chaubey T005
Average Accuracy: 97.33 %
```

B)Own dataset (Land.csv)

Code:-

```
import pandas as pd
from sklearn.model_selection import cross_val_score
from sklearn.neighbors import KNeighborsRegressor
# Load dataset
land = pd.read_csv("land.csv")
X = land[["Area"]]
y = land[["Price"]]
```

```
# Create regression model
model = KNeighborsRegressor(n_neighbors=2)
# Apply 3-fold cross-validation
scores = cross_val_score(model, X, y, cv=3, scoring="r2")
print("R2 scores for each fold:")
print(scores)
print("Average R2 Score:", round(scores.mean() * 100, 2), "%")
```

Output:-

```
R2 scores for each fold:
[-19.140625   -3.56790123 -13.2231405 ]
Average R2 Score: -1197.72 %
```

5. Implement a Decision Tree classifier using the Iris or any relevant dataset.**Code:-**

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
iris = pd.read_csv("iris - iris - iris - iris.csv")
X = iris.iloc[:, :-1]
y = iris.iloc[:, -1]
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.3,
    random_state=1
)
model = DecisionTreeClassifier()
model.fit(X_train, y_train)
print("Training Accuracy:", model.score(X_train, y_train) * 100)
print("Testing Accuracy:", model.score(X_test, y_test) * 100)
prediction = model.predict(X_test[0:1])
print("Predicted Class:", prediction[0])
print("Ritesh Chaubey T005")
```

Output:-

```
Training Accuracy: 100.0
Testing Accuracy: 95.55555555555556
Predicted Class: 0
Ritesh Chaubey T005
```

B)Own dataset(Land.csv)**Code:-**

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
# Load dataset
land = pd.read_csv("land.csv")
X = land[["Area"]]
y = land["Price"]
# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.3,
    random_state=1
)
# Create model
model = DecisionTreeClassifier()
# Train model
model.fit(X_train, y_train)
# Display accuracy
print("Training Accuracy:", model.score(X_train, y_train)*100)
print("Testing Accuracy:", model.score(X_test, y_test)*100)
# Prediction
prediction = model.predict(X_test[0:1])
print("Predicted Class:", prediction[0])
```

Output:-

```
Training Accuracy: 100.0
Testing Accuracy: 0.0
Predicted Class: 58
```